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Reinforced building wall using compression rod

Abstract

A reinforced building wall includes a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate and first and second vertical studs extending between the bottom plate and the top plate; a first threaded rod having a first end portion embedded in the foundation and a second end portion extending through the bottom plate; a compression rod having a first end portion operably attached to the second end portion of the first threaded rod and a second end portion operably bearing onto a bottom surface of the top plate, the compression rod is configured to handle compression forces from the top plate without buckling; an anchor embedded in the foundation, the anchor is operably attached to the first end portion of the first threaded rod.

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Background/Summary

RELATED APPLICATION
(1) This a nonprovisional application of Provisional Application Ser. No. 62/914,582, filed Oct. 14, 2019, hereby incorporated by reference, and is related to application Ser. No. 16/415,595, filed on May 17, 2019, hereby incorporated by reference.
FIELD OF THE INVENTION
(2) The present invention is generally directed to reinforced building walls designed to resist static and dynamic compression and tension forces.
BACKGROUND OF THE INVENTION
(3) Reinforced building walls using threaded rods anchored to the foundation are disclosed in the prior art. For example, see U.S. Pat. Nos. 6,951,078, 7,762,030, 8,136,318, 8,943,777, 9,097,000, 9,097,001, 9,416,530 and 9,874,009, hereby incorporated herein by reference. These walls are designed to hold the walls against tension loads or forces caused by earthquakes and/or high winds.
SUMMARY OF THE INVENTION
(4) The present invention provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate and first and second vertical studs extending between the bottom plate and the top plate; a first threaded rod having a first end portion embedded in the foundation and a second end portion extending through the bottom plate; a compression rod having a first end portion operably attached to the second end portion of the first threaded rod and a second end portion operably bearing onto a bottom surface of the top plate, the compression rod is configured to handle compression forces from the top plate without buckling; an anchor embedded in the foundation, the anchor is operably attached to the first end portion of the first threaded rod.
(5) The present invention also provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate and first and second vertical studs extending between the bottom plate and the top plate; a first threaded rod

having a first end portion embedded in the foundation and a second end portion extending through the bottom plate; a second threaded rod having a first end portion and a second end portion, the second end portion extending into an opening in the top plate; a compression rod having a first end portion operably attached to the second end portion of the first threaded rod and a second end portion operably attached to the first end portion of the second threaded rod; a first bearing plate disposed below and in contact with a bottom surface of the top plate and operably attached to the second end portion of the second threaded rod; and an anchor embedded in the foundation, the anchor is operably attached to the first end portion of the first threaded rod.
(6) The present invention further provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate, a vertical stud extending between the bottom plate and the top plate; a first threaded rod having a first end portion embedded in the foundation and a second end portion extending through the bottom plate; a compression rod having a first end portion operably attached to the second end portion of the first threaded rod and a second end portion operably attached to the top plate, the compression rod is configured to handle compression forces without buckling; a cross-member operably attached to the compression rod to transfer compression forces to the compression rod; the vertical stud including a bottom end supported by the cross-member to transfer compression forces to the cross-member; an anchor embedded in the foundation, the anchor is operably attached to the first end portion of the first threaded rod.
(7) The present invention provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate, a vertical stud extending between the bottom plate and the top plate; a first threaded rod having a first end portion embedded in the foundation and a second end portion extending through the bottom plate; a compression rod having a first end portion operably attached to the second end portion of the first threaded rod and a second end portion operably attached to the top plate, the compression rod is configured to handle compression forces without buckling and tension forces; a cross-member operably attached to the compression rod to transfer tension forces to the compression rod; the vertical stud including a top end supporting the cross-member to transfer tension forces to the cross-member; an anchor embedded in the foundation, the anchor is operably attached to the first end portion of the first threaded rod.
(8) The present invention further provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including first and second vertical studs supported by the foundation; a horizontal wood part supported by the first and second vertical studs; a compression rod having a first end portion operably supported by the foundation and a second end portion operably bearing onto a bottom surface of horizontal wood part.
(9) The present invention also provides a reinforced building wall, comprising a first stud wall disposed above a foundation; the first stud wall including a bottom plate, a top plate and first and second vertical studs extending between the bottom plate and the top plate; and a compression rod having a first end portion operably supported by the foundation and a second end portion operably bearing onto a bottom surface of the top plate.
(10) The present invention further provides an anchor embedded in a concrete structure, the anchor for attachment to a compression loading rod in a wood frame wall, comprising a threaded rod with first end portion for attaching to the compression loading rod and a second end portion embedded in the concrete structure; a first nut threaded to the threaded rod, the first nut engaging the concrete structure at an upper portion of the concrete structure; and the nut is configured to generate a shear cone in the concrete structure when the compression loading rod is subjected to compression forces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is perspective fragmentary view of a stud-framed building wall, incorporating an embodiment of the present invention.

(2) FIGS. **2-8** are enlarged perspective views of portions of the wall of FIG. **1**.

(3) FIG. **9** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(4) FIG. **10** is an enlarged perspective view of a portion of the wall of FIG. **9**.

(5) FIG. **11** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(6) FIGS. **12-13** are enlarged perspective views of portions of the wall of FIG. **11**.

(7) FIG. **14** is a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(8) FIG. **15** is an enlarged perspective view of a portion of the wall of FIG. **14**.

(9) FIG. **16** is a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(10) FIG. **17**A a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(11) FIGS. **17**B-**19** are enlarged perspective views of portions of the wall of FIG. **17**A, some shown with modifications.

(12) FIG. **20** a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(13) FIGS. **21-26** are perspective fragmentary views of a stud-framed building wall, incorporating modifications to the wall of FIG. **20**.

(14) FIG. **27** is an enlarged perspective view of a portion of the wall of FIG. **25**.

(15) FIG. **28** is a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(16) FIG. **29** is an enlarged perspective view of a portion of the wall of FIG. **28**.

(17) FIG. **30** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(18) FIG. **31** is an enlarged perspective view of a portion of the wall of FIG. **30**.

(19) FIG. **32** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(20) FIGS. **33-39**B are enlarged perspective views of portions of the wall of FIG. **32**, some shown with modifications.

(21) FIG. **40** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(22) FIGS. **41-45** are enlarged perspective views of portions of the wall of FIG. **40**, some shown with modifications.

(23) FIG. **46** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(24) FIGS. **47-48** are enlarged perspective views of portions of the wall of FIG. **46**.

(25) FIG. **49** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(26) FIGS. **50-53** are enlarged perspective views of portions of the wall of FIG. **50** with some shown with modifications.

(27) FIG. **54** is perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.

(28) FIGS. **55-57** are enlarged perspective views of portions of the wall of FIG. **54**, with some shown with modifications.

(29) FIG. **58** is a perspective fragmentary view of a stud-framed building wall, incorporating another embodiment of the present invention.
(30) FIGS. **59-62** are perspective fragmentary views of the stud-framed building wall of FIG. **58** with some modifications.
(31) FIGS. **63-64** are enlarged perspective views of portions of the wall of FIG. **58** with some shown with modifications.
(32) FIGS. **65**A-**69**B show the application of the present invention in walls supporting cross-laminated timber floorings.

DETAILED DESCRIPTION OF THE INVENTION

(33) The embodiments disclosed herein are configured to transfer compression or tension forces from horizontal or vertical framing members of a wall to the foundation or other concrete structure.
(34) Referring to FIG. **1**, a shear wall **2** incorporating the present invention is disclosed. The shear wall **2** forms a part of a building. The wall **2** is shown as a 2-story wall, comprising a lower wall **4**, and an upper wall **6**. It should be understood that the wall **2** may be one story or more than 2 stories, where one or more walls may be interposed between the lower wall **4** and the upper wall **6**.
(35) The wall **2** is supported on a concrete foundation **8**. It should be understood that the foundation **8** can also be a steel I-beam, wood beam, or other concrete structures such as concrete slab, concrete floor, concrete wall, or concrete column that is directly or indirectly anchored to the ground. The wall **2** is made of vertical studs **10**, which may be wood or metal. Wall sheathing (not shown) is operably attached to the studs **10**, such as with nails or screws. The studs **10** are operably attached to respective horizontal bottom plates **12** and horizontal top plates **14**. Floor joists **16** are supported on the respective top plates **14**. Subfloor **18** is supported by the floor joists **16**. Subfloor **18** is typically made of plywood sheets laid out over the floor joists **16**. The end studs **10**, while shown doubled up, may also be a single stud, depending on the expected load.
(36) The wall **2** is configured for tension and compression forces. Tension forces tend to lift the wall upwardly while compression forces tend to compress the wall downwardly. When one corner of the wall is being subjected to tension or lifting forces, the opposite corner of the wall is simultaneously subjected to compression forces.
(37) An anchor **20** is embedded in concrete in the foundation **8** in the foundation **8**. A compression rod **22** is operably attached to an anchor rod **24**. The anchor rod **24** is operably attached to the anchor **20**. The compression rod **22** is operably attached to an intermediate threaded rod **26** that extends through openings in in the top plate **14** and the bottom plate **12** of the wall **6** above. The rod **26** is attached to the top plate **14** and the bottom plate **12** with respective bearing plates **28** and nuts **30**. The bearing plate **28** may be made of solid or hollow metal.
(38) Several sections of the compression rod **22** may be joined end to end to form longer lengths to reach the topmost wall in a multi-story building, as disclosed discussed herein. The compression rod **22** is not limited to the lower wall but may extend to the upper walls to handle compression loadings in the upper walls. The compression rod **22** will also function as a tension rod to handle tension forces in a typical hold down system. The compression rod **22** when joined end to end, using intermediary threaded rods, reduced diameter threaded ends or couplings, as disclosed herein, will function as if made of one piece. See, for example, FIGS. **20**, **23**, **25**, **58**, **47**, **69**A.
(39) The compression rod **22** has sufficient cross-sectional area or strength or rigidity to handle compression forces without buckling. The compression rod **22** may also handle tension forces. The compression rod **22** may have the same or larger diameter than the threaded rod **26**, depending on the expected loading for the wall.
(40) Compression or downward forces are due to wall weight, shifting or racking of the wall **2** at the opposite end due to shear forces caused by earthquake or high winds. Tension or upward forces are due to lifting of the wall **2** due to shear forces from earthquake or high winds. For example, an earthquake typically generates tension and compression loading on the wall due to repeated racking of the wall from the ground motion. These compression and tension forces are dynamic loading

that typically occur only during these events so that designing the wood framing members to also handle these dynamic loadings would not be cost effective. The compression rod **22** advantageously replaces additional wood studs or posts that would be needed to handle the additional loading on the wall that otherwise would not be present most of the time. The compression rod **22** may be made of mild steel, which is cheaper than the high strength steel used for threaded rods in commercially available hold down systems. The use of the compression rod **22** advantageously reduces material and labor costs. The compression rod **22** may be solid or tubular and may be of any cross-sectional shape, such as circular, hexagonal, rectangular, etc.

(41) The lower bearing plate **28** attached to the bottom of the top plate **14** and the corresponding nut **30** advantageously transfer the compression forces from the horizontal framing member or top plate **14** to the compression rod **22** (see FIG. **4**). The upper bearing plate **28** on the bottom plate **12** and the corresponding nut **30** advantageously transfer the tension forces from the horizontal framing member or top plate **12** to the compression rod **22**.

(42) A tie rod **32** is joined to the intermediate threaded rod **26** with a coupling **34**. Another bearing plate **28** and a nut **30** are used to attach the tie rod **32** to the top plate **14** of the upper wall **6**. The tie rod **32** may be smaller in diameter than the compression rod **22**, depending on the expected load on the wall **6**. The tie rod **32** may be threaded only at the end portions, with the intermediate portion being smooth or unthreaded.

(43) Referring to FIG. **2**, the anchor rod **24** may be threaded at one end **36** and threaded to a threaded bore **38** at a bottom end of the compression rod **22**. The opposite end of the anchor rod **24** may also be threaded for attachment to the anchor **20** (shown in FIG. **1**). A sight hole **40** is provided for visually inspecting the extent of travel of the threaded end **36** into the threaded bore **38** to insure proper engagement of the anchor rod **24** with the compression rod **22**. The anchor rod **24** may also be threaded throughout its length as shown in FIG. **3**.

(44) Referring to FIG. **4**, the other end of the compression rod **22** includes another threaded bore **42** with a sight hole **44**. The intermediate threaded rod **26** is threaded to the threaded bore **42**. The intermediate portion of the intermediate threaded rod **26** may be unthreaded. Alternatively, the intermediate threaded rod **26** may be threaded throughout its length, as shown in FIG. **5**. The lower bearing plate **28** bearing onto the bottom surface of the top plate **14** is effective to transfer compression forces from the horizontal framing member or top plate **14** to the compression rod **22**, while the upper bearing plate **28** bearing on top of the horizontal framing member or bottom plate **12** is effective to transfer tension forces from the bottom plate **12** to the compression rod **22**.

(45) Referring to FIG. **6**, the opening **46** of the bearing plate **28** may be threaded for direct connection with intermediate threaded rod **26**. In this arrangement, the nut **30** may be omitted. T

(46) Referring to FIG. **7**, the threaded bore **42** may be multi-diameter with a larger threaded bore **48** and a smaller threaded bore **50**. The sight hole **44** may be associated with the larger threaded bore **48** and another sight hole **52** with the smaller threaded bore **50**.

(47) Referring to FIG. **8**, the threaded bore **38** may be multi-diameter with a larger threaded bore **54** and a smaller threaded bore **56**. The sight hole **40** may be associated with the larger threaded bore **54** and another sight hole **58** with the smaller threaded bore **56**. The bottom of the compression rod **22** bears on a bearing plate **28** to advantageously spreads a portion of the compression forces on the bottom plate **12**.

(48) It should be understood that each of the modifications shown in FIGS. **2**, **3** and **8** pertaining to the connection of the lower end portion of the compression rod **22** to the anchor rod **24** may be combined with each of the modifications shown in FIGS. **4-7** pertaining to the connection of the upper end portion of the compression rod **22** to the wall structure. Accordingly, the embodiment shown in FIG. **1** is understood to also include the various embodiments provided by combining each of the embodiments of FIGS. **2**, **3** and **8** with the embodiments of FIGS. **4-7**.

(49) Referring to FIGS. **9** and **10**, the embodiments provided by FIGS. **1-8** may be modified with the addition of an expandable connector **60** interposed between the bearing plate **28** and the nut **30**

on the bottom plate **12** and the top plate **14** of the upper wall **6**. The expandable connector **60** is well known in the art, as disclosed, for example, in U.S. Pat. Nos. 7,762,030, 6,161,350, 8,136,318, 8,186,924, 6,585,469, 2005/0055897, 2006/0156657, hereby incorporated herein by reference. The expandable connector **60** advantageously takes up any slack that develops in the intermediate threaded rod **26** and the tie rod **32** due to settlement or shrinkage of the wall **2**. The tie rod **32** may be threaded along its entire length. The connection of the compression rod **22** to the anchor rod **24** may be as shown for example in FIG. **2**, **3** or **8**.

(50) Referring to FIGS. **11-13**, the embodiments provided by FIGS. **1-8** may be modified with the elimination of the bearing plate **28** and the nut **30** on the bottom plate **12** of the upper wall **6**. Tension loading is carried by the bearing plate **28** on the top plate **14** in the upper wall **6**. The intermediate threaded rod **26** may be omitted and the tie rod **32** connected directly to the compression rod **22**. The tie rod **32** extends through oversized openings **62** in the top plate **14** and the bottom plate **12**. Compression (downward direction) forces on the tie rod **32** are advantageously transferred to the compression rod **22** through the bearing plate **28** and the nut **30**. The oversized openings **62** advantageously provides clearance to prevent or lessen the likelihood of the tie rod **32** getting caught and bowing as the wall moves relative to the tie rod **32** during uplift or downward motion of the wall caused by winds, storms, earthquake, etc. The compression rod **22**, which is shown as hexagonal or square, is also applicable to the embodiments of FIGS. **1-10** and all other embodiments disclosed herein using the compression rod **22**. The compression rod **22** may also be of any cross-sectional shape.

(51) Referring to FIGS. **14** and **15**, the embodiments of FIGS. **1-13** may be modified with the elimination of the tie rod **32** that connects to the top plate **14** of the upper wall **6**. The compression rod **22** is connected to the top plate **14** of the lower wall **4** and the bottom plate **12** of the upper wall **6** with the bearing plates **28** and the nuts **30**. The wall shown is designed for compression and tension forces. Tension forces from the upper wall **6** is advantageously transferred to the upper bearing plate **28** and compression forces on the lower bearing plate **28**.

(52) Referring to FIG. **16**, the embodiment of FIG. **14** is modified by attaching the upper bearing plate **28** to the top side of the top plate **14**. The compression rod **22** is connected to the top plate **14** of the lower wall **4** with a threaded rod **64** and the bearing plates **28** and the nuts **30**. The wall is configured to take compression forces through the lower bearing plate **28** and tension forces through the upper bearing plate **28**.

(53) Referring to FIGS. **17**A, **17**B and **18**, the embodiment of FIG. **16** is modified with the elimination of the upper bearing plate **28**. The threaded rod **64** is only connected to the underside of the top plate **14** to handle compression forces from the top plate **14**. The upper end of the threaded rod **64** is received within an opening **66** in the top plate **14**. Compression (downward direction) load from the wall is transferred to the compression rod **22** via the bearing plate **28** and the nut **30**. The bearing plate **28** may be attached to the top plate **14** with nails **68** or screws, as shown in FIG. **18**.

(54) The opening in the bearing plate **28** may be threaded, as shown in FIG. **6**, in which case the nut **30** is eliminated. The end of the rod **64** may penetrate the opening partway, or flush at the other side of the plate or past the opening. The opening may also be an unthreaded blind hole with the top end of the rod **64** bearing on the floor of the blind hole to transfer compression forces (downward direction) to the rod **64** and the compression rod **22**.

(55) Referring to FIG. **19**, the embodiment of FIG. **17**A is modified with the addition of a flanged sleeve **70** disposed in the opening **66**. The sleeve **70** advantageously provides a smooth opening through which the rod **64** can move through as the wall compresses without binding. The sleeve **70** includes a flange **72** that is supported by the top surface of the top plate **14** around the opening **66**. The sleeve **70** may also be threaded to the rod **64** to resist tension loading (uplift forces) via the flange **72** bearing on the top plate **14**.

(56) Referring to FIGS. **20-27**, the compression rod **22** shown in FIGS. **1-19** may be modified into

two pieces—compression rod **74** and compression rod **76**—joined end to end. This advantageously provides for easier handling during installation as each of the compression rods **74** and **76** would weigh less than a single compression rod **22** and would be shorter in length, providing for easier handling.

(57) Referring to FIG. **20**, the top edge **80** of the compression rod **74** is spaced from the bottom edge **82** of the compression rod **76**. Each of the compression rods **74** and **76** is made the same as the single compression rod **22** shown in FIGS. **1-19**. The threaded bores **38** and **42** may be single diameter or multi-diameter as shown in FIGS. **7** and **8**.

(58) Referring to FIG. **21**, the joint between the compression rods **74** and **76** shown in FIG. **20** may be modified by using a shorter threaded rod **78** so that the top edge **80** of the compression rod **74** and bottom edge **82** of the compression rod **76** bear on each other when the threaded rod **78** is threaded to the threaded bores **38** and **72**. In this arrangement, the compression forces from the upper compression rod **76** are transferred to the lower compression rod **74** via the threaded engagement of the threaded rod **78** with the compression rods **74** and **76** and the bearing engagement of the top and bottom edges **80** and **82**.

(59) Referring to FIGS. **22** and **23**, the joint between the compression rods **74** and **76** shown in FIG. **20** may be modified by providing the upper end portion **84** of the compression rod **74** with a reduced diameter, outside threaded portion **86** for being threaded to the threaded bore **38** of the upper compression rod **76**. The end portion **84** has circumferential flange **88** that bears against the bottom edge **82** of the compression rod **76**. The sight hole **40** advantageously insures sufficient engagement of the threaded portion **86** inside the threaded bore **38**. The compression forces from the upper compression rod **76** are transferred to the lower compression rod **74** via the threaded engagement of the threaded portion **86** with the compression rod **76** and the bearing engagement of the surfaces **82** and **88**.

(60) Referring to FIGS. **24** and **25**, the joint between the compression rods **74** and **76** shown in FIG. **20** may be modified by providing the upper end portion **84** of the compression rod **74** and the lower end portion **90** of the compression rod **76** with outside threads **92** and connecting the two ends together with a coupling **94**, which is preferably cylindrical with a longitudinal slot **96** that provides an inspection opening for the top surface **98** of the lower compression rod **74** and the bottom surface **100** of the upper compression rod **76**. The slot **96** advantageously provides a way for visually checking the extent of travel of the threads **92** inside the coupling **94**. The surfaces **98** and **100** are preferably engaging each other inside the coupling **94**. The slot **96** is preferably centered on the length of the coupling **94**.

(61) Referring to FIGS. **26** and **27**, the compression rods **74** and **76** shown in FIGS. **24** and **25** may be tubular joined by the coupling **94**. The abutting ends of the compression rods **74** and **76** may also be connected using the configuration shown in FIG. **21**, where the abutting ends are provided with threaded bores **38** and **42** and joined to a threaded rod **78**. The opposing ends of the compression rods **74** and **76** may be provided with inside threads and joined in the manner shown in FIGS. **20** and **21** with the threaded rod **78**.

(62) It should be understood that the compression rods **74** and **76** may be attached to the foundation and the wall structure in the same way as already disclosed above, as for example shown in FIGS. **1-19**.

(63) Referring to FIGS. **28** and **29**, the compression rod **22** shown in FIGS. **1-19** or the two-piece compression rods **74** and **76** shown in FIGS. **20-27** may be threaded throughout their entire length.

(64) Referring to FIGS. **30** and **31**, the compression rod **22** shown in FIGS. **1-19** or the two-piece compression rods **74** and **76** shown in FIGS. **20-27** may be modified where the end portions **102** and **104** are provided with respective reduced diameter, outside threaded portions **86** connected to the anchor rod **24** and the intermediate threaded rod **26** with the respective couplings **34**.

(65) Referring to FIGS. **32**, **33** and **34**, the connection of the upper end of the compression rod **22** to the top plate **14** shown in FIGS. **1-31** may be modified by providing the end portion **102** of the

compression rod **22** with an outside thread **106** in addition to the inside thread **108**. A cylindrical coupling **110** connects the compression rod **22** to the intermediate threaded rod **26**. The coupling **110** bears against the bearing plate **28**. The sight hole **40** provides a visual check on the extent of travel of the threads **106** inside the coupling **110**. The nut **30** used in connection with the bearing plate becomes unnecessary with the use of the coupling **110**. Referring to FIG. **34**, the cylindrical coupling **110** may be modified with a circumferential flange **112** to take the place of the bearing plate **28**. The compression rod **22** may be solid or tubular.
(66) Referring to FIG. **35**, the anchor **20** shown in the various embodiments of the wall, for example in FIG. **1**, includes an anchor body **114** supported on a support or chair **116**. The anchor rod **24** is supported by the chair **116**. The anchor body **114** is threaded to the anchor rod **24**. The anchor body **114** may be a standard nut, a metal plate, a cylindrical body, or any of the anchors disclosed in U.S. Pat. Nos. 8,943,777, 9,097,001, 9,222,251, 9,416,530, 9,447,574, 9,702,139, 9,874,009, hereby incorporated herein by reference. The chair **116** may be any support that positions the anchor body **114** a distance from the bottom of the **118** of the concrete foundation **8**. Examples of the chair **116** are disclosed in U.S. Pat. Nos. 9,222,251 and 8,943,777, hereby incorporated herein by reference.
(67) Referring to FIG. **36**, the anchor **20** shown in FIG. **35** may be modified with the addition of an upper anchor body **114** disposed above the lower anchor body **114**. The upper anchor body **114** is configured to generate a shear cone **120** in the concrete foundation **8** when subjected to compression (downward direction) forces. Similarly, the lower anchor body **114** is configured to generate a shear cone **122** when subjected to tension (upward direction) forces. Accordingly, the anchor arrangement shown can transfer both compression and tension forces to the foundation **8**. The upper anchor body **114** is shown embedded flush with the upper surface **124** of the foundation **8** but it should be understood that it may be disposed below or on top of the top surface **124**, depending on the expected load and the volume of the shear cone **120** sufficient to support the load.
(68) Referring to FIG. **37**, the upper anchor body **114** shown in FIG. **36** may be a nut **30**. The embodiment of FIG. **36** may be further modified by providing a cross-member or compression plate **126** disposed over the bottom plate **12** to support the bottom ends of two vertical studs **10**. The compression plate **126** has a threaded opening **128**, which is threaded to the anchor rod **24**. In this configuration, compression forces from the vertical framing members or studs **10** are transferred to the rod **24** through the cross-member **126**. An additional advantage of the plate **126** is that it has a higher compressive strength than the underlying bottom plate **12** so that forces on the compression plate **126** is spread over a larger area on the bottom plate **12**, as disclosed in co-pending application Ser. Nos. 16/296,865 and 16/415,595, hereby incorporated herein by reference. The plate **126** is preferably made of metal.
(69) Referring to FIG. **38**, the embodiment of FIG. **37** may be modified by disposing the nut **30** on the top surface **124** of the concrete foundation **8**. An opening **130** in the bottom plate **12** receives the nut **30**.
(70) Referring to FIG. **39**A, the embodiment of FIG. **37** may be modified by raising the compression plate **126** above the bottom plate **12**, creating a gap **132** between the compression plate **126** and the bottom plate **12**. The nut **30** is embedded in the foundation and flush with the top surface **124** of the concrete foundation **8**. The compression plate **126** is threaded to the anchor rod **24**, advantageously allowing compression loading from the vertical framing members or studs **10** bearing on the plate **126** to be transferred directly to the rod **24** via the cross-member **126**. The plate **126** is preferably made of metal.
(71) Referring to FIG. **39**B, the embodiment of FIG. **39**A may be modified with the addition of an upper nut **30** within the gap **132**. The compression plate **126** is provided with a non-threaded opening **133**. Compression loading from the vertical framing members or studs **10** bearing on the plate **126** is transferred to the rod **24** via the plate **126** bearing on the upper nut **30**. The upper nut **30** disposed on the bottom plate **12** engages the bottom portion of the plate around the opening **133**.

(72) Referring to FIGS. **40** and **41**, the embodiment shown in FIG. **9** may be modified by combining the embodiment of FIG. **37** and the provision of reinforcement studs **136** that extend between the compression plate **126** and a cross-member **138**. Reinforcement studs **140** also extend between the cross-member **138** and the top plate **14**. The reinforcement studs **136** and **140** are operably attached to the respective adjacent studs **10** by nails or screws. The bottom ends of the reinforcement studs **136** bear on the bearing plate **126**. The cross-member **138** is supported by the top ends of the reinforcement studs **136**. The bottom ends of the reinforcement studs **140** bear on the cross-member **138**. The threaded rod **26** extends through an unthreaded opening **142** in the cross-member **138** and is threaded to the threaded bore **42**. The nut **30** secures the cross-member **138** against the top edge **144** of the compression rod **22**. In this configuration, compression and tension forces from the vertical framing members or reinforcement studs **136** and **140** are transferred to the compression rod **22** via the cross-member **138**. When the wall is under compression, the studs **140** transfer compression forces to the compression rod via the cross-member **138**. When the wall is under tension, the studs **136** transfer tension forces to the compression rod **22** via the cross-member **138**.
(73) Referring to FIG. **42**, the embodiment of FIG. **40** may be modified wherein the reinforcement studs **140** are omitted. In this configuration, tension forces from above are transferred to the compression rod **22** via the threaded rod **26** and to the reinforcement studs **136** via the cross-member **138**, which is held in place by the nut **30**. Compression forces from the top plate **14** are transferred to the compression rod **22** via the bearing plate **28** and the threaded rod **26**.
(74) Referring to FIG. **43**, the embodiments of FIGS. **41** and **42** may be modified by providing the cross-member **138** with a threaded hole **146**, eliminating the use of the nut **30** to attach the cross-member **138** to the compression rod **22**.
(75) The cross-member **138** may be solid metal as shown in FIGS. **40-43** or hollow metal **148** with longitudinal openings **150** as shown in FIG. **44**, and disclosed in U.S. Pat. No. 9,097,000, hereby incorporated herein by reference.
(76) The cross-member **138** may be a solid wood member **152**, as shown in FIG. **45**.
(77) It should be understood that the cross-members **138**, **148** and **152** are interchangeable.
(78) Referring to FIGS. **46-48**, the embodiment of FIG. **40** may be modified wherein the compression rod **22** is replaced with the two-piece compression rods **74** and **76** shown in FIG. **20** and wherein the cross-member **138** is disposed between the opposing ends of the compression rods **74** and **76**.
(79) Referring to FIG. **47**, the upper and lower compression rods **76** and **74** are joined with the threaded rod **78** threaded to the threaded bores **42**. The opening **142** is preferably unthreaded.
(80) Referring to FIG. **48**, the cross-member **138** is provided with the threaded hole **146**. The threaded rod **78** is threaded to the hole **146**.
(81) Referring to FIGS. **49** and **50**, the embodiment shown in FIG. **43** may be modified wherein the end portion of the compression rod **22** is provided with a smaller diameter portion **154** with a circumferential shoulder **156** which provides support to the cross-member **138**. The opening **142** is preferably unthreaded.
(82) Referring to FIG. **51**, the embodiment of FIG. **49** may be modified wherein the reinforcement studs **136** shown in FIG. **50** are removed. In this configuration, the cross-member **138** can transfer compression forces from the vertical framing members or reinforcement studs **140** to the compression rod **22**. In addition, compression forces from the horizontal framing member or top plate **14** are transferred to the bearing plate **28** and thence to the compression rod **22**.
(83) Referring to FIGS. **52** and **53**, the embodiment of FIG. **42** may be modified wherein the top end **158** of the compression rod **22** is spaced apart from the cross-member **138**. The threaded rod **26** is threaded to the threaded bore **42** and the threaded hole **146**. In this configuration, compression forces from above are transferred to the compression rod **22** via the bearing plate underneath the top plate **14** and the threaded to the rod **26**. Similarly, tension forces from the reinforcement studs

136 are transferred to the compression rod **22** via the threaded rod **26** and the cross-member **138**, which is threaded to the rod **26**.
(84) Referring to FIGS. **54** and **55**, the embodiment shown in FIG. **1** may be modified wherein one end of each of the bearing plates **28** is extended to encompass the top ends **160** and the bottom ends **162** of the respective studs **10**. The extended bearing plate **28**, preferably made of solid metal, has a larger compressive strength than the top plate **14** and the bottom plate **12** to minimize crushing the horizontally oriented wood fibers of the top plate and the bottom plate by the vertically oriented wood fibers of the studs **10**. Since the studs **10** with their vertically oriented wood fibers are able to carry more load than the top plate and the bottom plate with their horizontally oriented wood fibers, the bearing plate **28** advantageously spread the forces from the top ends **160** and bottom ends **162** of the studs **10** into a larger area on the respective top plate **14** and the bottom plate **12**. The extended bearing plate **28** preferably has an elongated opening **163** to advantageously provide adjustability to the positioning of the bearing plate **28**. Co-pending application Ser. No. 16/296,865, hereby incorporated herein by reference, discloses the use of bearing plate **28** as a compression plate to reduce crushing of the underlying wood part.
(85) In the configuration shown in FIG. **55**, the lower extended bearing plate **28** underneath the top plate **14** is able to transfer compression forces from the horizontal framing member **14** to the compression rod **22**. The extended bearing plate **28** advantageously provides the additional function of the cross-member or compression plate **126**. In the case where the upper extended bearing plate on top of the bottom plate **12** is threaded to the rod **26**, the extended bearing plate **28** is able to transfer compression forces from the vertical framing members or studs **10** to the compression rod **22**.
(86) Referring to FIG. **56**, the extended bearing plate **28** shown in FIG. **55** may be further extended to encompass the top end **164** and the bottom end **166** of the opposite studs **10**.
(87) Referring to FIG. **57**, the embodiment of FIG. **56** may be modified wherein a bearing plate **28** on top of the top plate **14** is added for tension forces.
(88) Referring to FIG. **58**, the tie rod **32** in the upper wall **6** shown in the configuration of FIG. **1** is replaced with a compression rod **22**. A bearing plate **28** is provided on the underside of the top plate **14** of the upper wall **6**. The additional bearing plate **28** provides for transfer of compression forces to the upper compression rod **22**.
(89) Referring to FIG. **59**, the embodiment of FIG. **58** may be modified wherein the expandable connector **60** is interposed between the bearing plate **28** and the nut **30** on the bottom plate **12** and the top plate **14** of the upper wall **6**.
(90) Referring to FIG. **60**, the embodiment of FIG. **58** may be modified wherein the bearing plate **28** on the bottom plate **12** of the upper wall **6** is eliminated. Compression forces are transferred to the compression rods **22** through the bearing plates disposed on the respective underside of the top plates **14**. Tension forces are transferred to the compression rods **22** through the bearing plate **28** on the top plate **14** of the upper wall **6**.
(91) Referring to FIG. **61**, the embodiment of FIG. **60** may be modified wherein the bearing plate **28** underneath the top plate **14** of the lower wall **4** is eliminated. Compression forces are transferred to the compression rods **22** through the bearing plate **28** underneath the top plate **14** of the upper wall **6**. Tension forces are transferred through the bearing plate **28** on the top side of the top plate **14** of the upper wall **6**.
(92) Referring to FIG. **62**, the embodiment of FIG. **58** may be modified wherein the lower compression rod **22** in the lower wall **4** is not directly connected to the upper compression rod **22** in the upper wall **6**. The lower compression rod **22** is connected to the top plate **14** with a threaded rod **168**. Bearing plates **28** are attached to the rod **168** with nuts **30**. The upper compression rod **22** is attached to the bottom plate **12** and the top plate **14** of the upper wall **6** through the bearing plates **28** and the nuts **30**.
(93) Compression and tension forces from the upper wall **6** are transferred to the lower

compression rod **22** through the subfloor **18**, the floor joists **16**, the top plate **14** and the wall sheathing (not shown, typically plywood, oriented strand board, etc.) that is nailed on the outside of the studs **10**.

(94) Referring to FIG. **63**, the expandable connector **60** is of the type shown in U.S. Pat. No. 8,136,318, with an inner cylindrical member **168** axially movable within an outer cylindrical member **170** under the action of a spring **172**. The bottom end **174** of the upper compression rod **22** engages the inner cylindrical member **168** so that any slack that may develop in the intermediate threaded rod **26** is taken up by the expandable connector **60** as the inner cylindrical member **168** moves upwardly while the outer cylindrical member **170** remains engaged with the bearing plate **28**.

(95) Referring to FIG. **64**, the expandable connector **60** is of the type disclosed in U.S. Pat. No. 10,337,185, (FIGS. 41-43) hereby incorporated herein by reference. This type of expandable connector works by relative rotational movement between an outer cylindrical member **176** and inner cylindrical member **178** due to the unwinding of a torsion spring **180**. The rotational movement serves to expand the connector to take up any slack in the rod **26**. The bottom end **174** of the upper compression rod **22** engages the inner cylindrical member **178** so that any slack that may develop in the intermediate threaded rod **26** is taken up by the expandable connector **60**. The clip **182** is removed to activate the connector.

(96) Referring to FIGS. **65**A and **65**B, a wall **184** using a cross-laminated timber (CLT) flooring **186** is disclosed. Use of CLT flooring **186** is disclosed in application Ser. Nos. 16/415,595 and 16/780,472, hereby incorporated by reference. The compression rod **22** is designed to resist compression forces from the CLT flooring **186**. The bearing plate **28** transfers the compression forces from the CLT flooring **186** to the compression rod **22** and the foundation **8**. Single studs **10** bearing on the top surface of the foundation support the static load of the flooring **186**. The compression rod **22** supports compression forces during dynamic loading caused by shear forces on the wall **184** during earthquakes, hurricanes, storms, tornadoes, etc. The bottom end of the compression rod **22** bears on the top surface of the foundation **8**. The top end of the compression rod **22** is operably attached to the bearing plate **28**. Referring to FIG. **65**B, the compression rod **22** is threaded to the bearing plate **28** in the opening **46** with the end of the compression rod being flush with top side of the bearing plate **28**.

(97) Referring to FIGS. **66**A-**66**B, the wall embodiment of FIG. **65**A may be modified wherein the bottom end portion of the compression rod **22** is embedded in the concrete foundation **8** with a nut **30** bearing on the top surface of the foundation **8**.

(98) Compression forces from the CLT flooring **186** are transferred to the compression rod **22** through the bearing plate **28**, which is attached to the compression rod **22** with a nut **30** or a threaded anchor body **114**, or any threaded body such as a metal plate, a cylindrical body, or any of the anchors disclosed in U.S. Pat. Nos. 8,943,777, 9,097,001, 9,222,251, 9,416,530, 9,447,574, 9,702,139, 9,874,009. The opening **46** in the bearing plate **28** is unthreaded. The top end portion of the compression rod **22** may extend into the opening **46** partway. Single or double studs **10**, depending on the load in the bottom floor in the stud bay in which the compression rod **22** is disposed, may be used to support the static loading, with the compression rod **22** providing additional support for the dynamic loading during earthquakes, hurricanes, storms, tornadoes, etc.

(99) Referring to FIG. **67**, the compression rod **22** shown in FIG. **65**A may be modified to support compression forces from an upper CLT flooring **186**, which may be one or more stories above the ground floor. The top end portion of the compression rod **22** is shown threaded to the bearing plate **28** but an unthreaded opening **46** and a nut **30**, as shown in FIG. **66**B may also be used. The compression rod **22** extends through an opening in the lower CLT flooring **186** and ends at the bearing plate **28** underneath the upper CLT flooring **186**. The depth of the opening in the lower CLT flooring **186** advantageously helps stabilize the compression rod **22** when under compression. Single or double studs **10**, depending on the load in the bottom floor, may be used to support static

loading.
(100) Referring to FIG. **68**, the embodiment shown in FIG. **67** may be modified wherein tension and compression loadings from the lower CLT flooring **186** are added to the compression rod **22**. The bearing plate on top of the lower CLT flooring **186** transfers tension loading to the compression rod **22**. The bearing plate **28** on the underside of the lower CLT flooring **186** provides compression loading from the lower CLT flooring **186** to the compression rod **22**. The compression rod **22** is shown threaded to the bearing plates **28** but the bearing plates may also use unthreaded openings **46** in combination with the nuts, as shown in FIG. **66**B, to attach the bearing plates to the compression rod **22**.
(101) To handle the tension forces, the anchor **20** using a nut **30** is embedded in a lower portion of the concrete foundation **8** (see FIG. **36**). The nut **30** embedded in the upper portion of the concrete foundation **8** is used to transfer compression forces to the foundation. The nut **30** for transferring the compression forces to the foundation may be disposed on top of the concrete foundation (FIG. **66**A) or embedded flush or below the top surface of the foundation in the upper portion of the concrete foundation. The nut **30** may be substituted with the threaded anchor body **114**, or any threaded body such as a metal plate, a cylindrical body, or any of the anchors disclosed in U.S. Pat. Nos. 8,943,777, 9,097,001, 9,222,251, 9,416,530, 9,447,574, 9,702,139, 9,874,009.
(102) Referring to FIGS. **69**A and **69**B, the compression rod **22** may be made of several sections joined with couplers to achieve the required length. For example, the lower portion **188** of the compression rod **22** may be a separate piece joined to the rest of the compression rod **22** with the coupling **34**. Along with the nut **30**, the lower portion **188** becomes a bolt (headless) and nut assembly that is used for anchoring the compression rod **22** to the concrete foundation **8**.
(103) It should be understood that the various components shown in a particular configuration of a wall may be interchanged with other components shown in another configuration of a wall to arrive at a different configuration to provide a wall reinforced for tension and/or compression forces. The wall shown is reinforced for compression and tension forces. The lower bearing plate **28** below the upper CLT flooring **186** transfers compression loading to the compression rod **22** to the concrete foundation **8** via the upper nut **30**. The upper bearing plate **28** on top of the upper CLT flooring **186** transfers tension loading to the compression rod **22** to the concrete foundation **8** via the lower nut **30**.
(104) Referring to FIG. **69**B, the upper nut **30** generates a shear cone **120** in the concrete foundation **8** in response to compression forces on the compression rod **22**. Similarly, the lower nut **30** generates a shear cone **122** in response to tension forces on the compression rod **22**. See also FIG. **36**.
(105) The configurations of the compression rods **22** shown in FIGS. **65**A-**69**B are equally applicable to stud walls not using the CLT floorings **186**, such as the wall shown in FIG. **1**.
(106) It should also be understood that where double studs are shown in a particular configuration of a stud wall, a single stud is equally applicable, depending on the expected load.
(107) While this invention has been described as having preferred design, it is understood that it is capable of further modifications, uses and/or adaptations following in general the principle of the invention and including such departures from the present disclosure as come within known or customary practice in the art to which the invention pertains, and as may be applied to the essential features set forth, and fall within the scope of the invention or the limits of the appended claims.

Claims

1. A reinforced building wall, comprising: a) a first stud wall disposed above a foundation; b) the first stud wall including a first bottom plate and a first top plate; c) a second stud wall disposed above the first stud wall, the second stud wall including a second bottom plate and a second top plate; d) a compression rod having a first end with a first threaded bore and a second end with a

second threaded bore, the first threaded bore and the second threaded bore being formed in and monolithic with the respective first end and the second end of the compression rod, the first end being operably attached to the foundation; e) a horizontal first member disposed in and operably attached to the first stud wall, the first member being operably attached to the compression rod to transfer compression forces to the compression rod and the foundation; and f) a horizontal second member operably attached to the second stud wall and operably attached to the compression rod to transfer tension forces to the compression rod.

2. The reinforced building wall as in claim 1, wherein the first member includes a first bearing plate engaging an underside of the first top plate.

3. The reinforced building wall as in claim 2, wherein: a) a first threaded rod is operably attached to the second threaded bore of the compression rod; and b) a first fastener is threaded to the first threaded rod and bears on an underside of the first bearing plate.

4. The reinforced building wall as in claim 3, wherein: a) a second bearing plate is disposed on the first top plate; and b) a second fastener is threaded to the threaded rod and bears on the second bearing plate.

5. The reinforced building wall as in claim 2, wherein: a) a threaded rod is operably attached to the second threaded bore of the compression rod; and b) the first bearing plate is threaded to the threaded rod.

6. The reinforced building wall as in claim 2, wherein the first threaded bore and the second threaded bore include respective sight holes.

7. The reinforced building wall as in claim 6, wherein at least one of the first threaded bore and the second threaded bore includes multi-diameter threads.

8. The reinforced building wall as in claim 2, wherein a coupling is operably attached to the compression rod and bears on the first bearing plate.

9. The reinforced building wall as in claim 2, wherein a first stud includes a first top end engaging an underside portion of the first bearing plate.

10. The reinforced building wall as in claim 9, wherein: a) a threaded rod is operably attached to the second threaded bore of the compression rod; and b) a first fastener is threaded to the threaded rod and bears on the first bearing plate.

11. The reinforced building wall as in claim 9, wherein a second stud includes a second top end engaging another underside portion of the first bearing plate.

12. The reinforced building wall as in claim 1, wherein the first threaded bore is operably attached to a first threaded rod anchored in the foundation.

13. The reinforced building wall as in claim 12, wherein: a) a bearing plate is disposed on the first bottom plate; and b) the first threaded rod extends through the first bottom plate and the bearing plate.

14. The reinforced building wall as in claim 13, wherein a first stud includes a first bottom end bearing on one end the bearing plate.

15. The reinforced building wall as in claim 14, wherein a second stud includes a second bottom end bearing on an another end of the bearing plate.

16. The reinforced building wall as in claim 1, wherein the first member includes a bearing plate disposed on the first bottom plate.

17. The reinforced building wall as in claim 1, wherein: a) a coupling is attached to the compression rod; and b) the coupling includes a flange portion engaging an underside of the first top plate.

18. The reinforced building wall as in claim 1, wherein the first member includes a cross member operably attached to the second end of the compression rod and operably attached to the first stud wall.

19. The reinforced building wall as in claim 18, wherein: a) a threaded rod is operably attached to the second threaded bore of the compression rod; and b) a first fastener is threaded to the threaded

rod and bears on the first member.
20. The reinforced building wall as in claim 19, wherein: a) a first bearing plate engages an underside of the first top plate; and b) a second fastener is threaded to the threaded rod and bears on the first bearing plate.
21. The reinforced building wall as in claim 18, wherein: a) a threaded rod is operably attached to the second threaded bore of the compression rod; and b) the threaded rod is threaded to the first member.
22. The reinforced building wall as in claim 18, wherein a first stud and a second stud include top ends engaging an underside of the first member.
23. The reinforced building wall as in claim 22, wherein a third stud and a fourth stud include bottom ends disposed on top of the first member.
24. The reinforced building wall as in claim 18, wherein the first member is hollow metal.
25. The reinforced building wall as in claim 18, wherein the first member includes wood.
26. The reinforced building wall as in claim 18, wherein the first member is supported on the second end of the compression rod.
27. The reinforced building wall as in claim 26, wherein the compression rod includes a reduced diameter portion disposed inside an opening in the first member.
28. The reinforced building wall as in claim 1, wherein the compression rod includes a first section and a second section operably joined end-to-end with a first free end having the first threaded bore and a second free end having the second threaded bore.
29. The reinforced building wall as in claim 28, wherein the first section and the second section include respective opposite threaded bores and a threaded rod is threaded to the opposite threaded bores.
30. The reinforced building wall as in claim 29, wherein the first member is disposed between the first section and the second section.
31. The reinforced building wall as in claim 30, wherein the threaded rod is threaded to the first member.
32. The reinforced building wall as in claim 29, wherein the opposite threaded bores include respective sight holes.
33. The reinforced building wall as in claim 28, wherein the first section is threaded into a threaded bore in the second section.
34. The reinforced building wall as in claim 28, wherein a coupling joins the first section to the second section.
35. The reinforced building wall as in claim 34, wherein the first section and the second section are tubular.
36. The reinforced building wall as in claim 28, wherein the first section is in bearing engagement with the second section.
37. The reinforced building wall as in claim 1, wherein a horizontal third member is operably attached to the compression rod to transfer compression forces to the compression rod.
38. The reinforced building wall as in claim 37, wherein: a) a threaded rod is operably attached to the second threaded bore of the compression rod; b) the threaded rod is threaded to the third member; and c) a first stud and a second stud include top ends engaging an underside of the third member.
39. The reinforced building wall as in claim 1, wherein: a) the second member includes a bearing plate disposed on the second bottom plate; and b) a threaded rod is operably attached to the second threaded bore of the compression rod; and c) a first fastener is threaded to the threaded rod and bears on the second member.
40. The reinforced building wall as in claim 39, wherein a first stud includes a first bottom end disposed on top of the second member.
41. The reinforced building wall as in claim 40, wherein a second stud includes a second bottom

end disposed on top of the second member.

42. The reinforced building wall as in claim 1, wherein: a) the second member includes a bearing plate disposed on the second top plate; b) a threaded rod is operably attached to the second threaded bore of the compression rod; and c) a first fastener is operably threaded to the threaded rod and bears on the second member.

43. The reinforced building wall as in claim 1, wherein: a) the second member includes a bearing plate disposed on the first top plate; b) a threaded rod is operably attached to the second threaded bore of the compression rod; and c) a first fastener is operably threaded to the threaded rod and bears on the second member.

44. The reinforced building wall as in claim 1, wherein: a) the compression rod is operably attached to a first threaded rod anchored to the foundation; and b) a first anchor is attached to the first threaded rod.

45. The reinforced building wall as in claim 44, wherein the first anchor includes a surface aligned with a top surface of the foundation.

46. The reinforced building wall as in claim 45, wherein the surface of the first anchor is disposed on the top surface of the foundation.

47. The reinforced building wall as in claim 44, wherein a second anchor is attached to the first threaded rod below the first anchor.

48. The reinforced building wall as in claim 47, wherein the first anchor and the second anchor each includes a nut.

49. A reinforced building wall, comprising: a) a first stud wall disposed above a foundation, the first stud wall including a first bottom plate and a first top plate; b) a second stud wall disposed above the first stud wall, the second stud wall including a second bottom plate and a second top plate; c) a compression rod having a first end with a first threaded bore and a second end with a second threaded bore, the first threaded bore and the second threaded bore being formed in and monolithic with the respective first end and the second end of the compression rod, the first end being operably attached to the foundation; d) a cross member operably attached to the compression rod to transfer compression forces to the compression rod and the foundation; and e) a bearing plate on the second top plate and operably attached to the compression rod to transfer tension forces to the compression rod.

50. A reinforced building wall, comprising: a) a first stud wall disposed above a foundation, the first stud wall including a first bottom plate and a first top plate; b) a second stud wall disposed above the first stud wall, the second stud wall including a second bottom plate and a second top plate; c) a compression rod having a first end with a first threaded bore and a second end with a second threaded bore, the first threaded bore and the second threaded bore being formed in and monolithic with the respective first end and the second end of the compression rod, the first end being operably attached to the foundation; d) a first bearing plate engaging an underside of the first top plate and operably attached to the compression rod to transfer compression forces to the compression rod and the foundation; and e) a second bearing plate disposed on the second top plate and operably attached to the compression rod to transfer tension forces to the compression rod.